

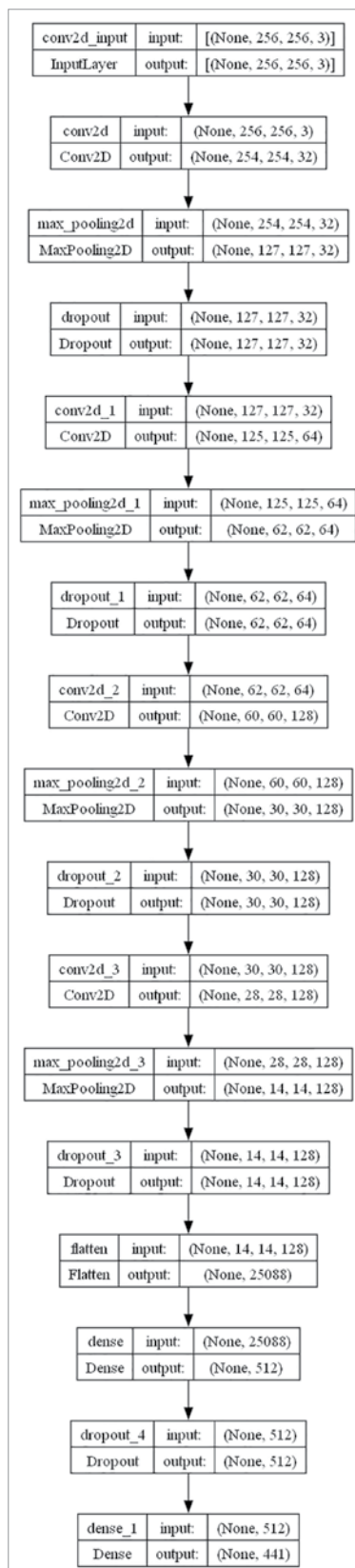


Figure 2: Model layers

```
normalized, sr=sr, x_axis='time', y_
axis='mel')
plt.savefig("/mnt/d/kaggle/
birdchef-images/" + dirval + "/" +
file.replace("ogg", "png"), bbox_
inches='tight', pad_inches=0)
plt.close()
```

These save the sound as an image.

## The layer architecture of the model

Figure 2 depicts the model layer architecture developed using Python and the open source libraries listed below.

- **Librosa:** Used for feature extraction of the Mel spectrogram.
- **Matplotlib:** Used to create visualisations of Mel frequencies, which are extracted from audio signals using the librosa library. It allows customisable and interpretable plots. This plotted data is saved as images.
- **TensorFlow and Keras:** Used for training the image data using a CNN.
- **NumPy:** Used for predictions.

The different layers of the model are convolution layer, pooling layer, dropout layer and connected layers.

The important features of the image are extracted in the convolution layer. The convolution operation involves applying a filter to the image, which produces a feature map. The filter is a small matrix of weights

that is learned by the model during training. The feature map shows how well the filter matches different parts of the image.

The pooling layer is used to reduce the dimensionality of the data. Pooling involves down-sampling the feature maps by taking the maximum or average value of a sub-region.

The dropout layer is used to prevent the model from overfitting to the training data. Dropout works by randomly dropping out a certain percentage of units from the neural network during training.

Connected layers are used to classify the image into one of several categories. This layer takes the output of the pooling layer and flattens it into a vector. The vector is then multiplied by a weight matrix and added to a bias vector. This produces a score and probabilities for each category. The category with the highest probability is the predicted class of the image.

A bird voice recognition model is a valuable tool for ornithology and conservation, especially in biodiverse areas. It helps in bird identification, bird population monitoring, habitat usage and migrant monitoring. Bird voice recognition enhances the understanding of avian biodiversity and ecological changes, and supports informed conservation. Protecting bird populations and habitats is vital for promoting a balanced environment. **END** 🐧

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